SCHOOL OF COMPUTATION,
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TECHNISCHE UNIVERSITÄT MÜNCHEN

Master's Thesis in Informatics

Vision-Language Foundation Models for Multi-Modal Medical Data

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Vision-Language-Foundation-Modelle für Multimodale Medizinische Daten

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I confirm that this master's thesis is my own work and I have documented all sources and material used.

Munich, 16.03.2026 Muhammed Göktepe

Acknowledgments

Abstract

Contents

Acknowledgments **iii**

Abstract **iv**

1 Introduction **1**
1.1 Problem Statement 1
1.2 Thesis Formulation 2
1.3 Contributions 2

2 Related Work **3**
2.1 Vision Transformers in Medical Imaging 3
2.1.1 Architectural Evolutions: Hierarchical and Self-Supervised ViTs 3
2.1.2 Adaptation to 3D Medical Volumes 4
2.2 Vision–Language Models 5
2.2.1 Contrastive Alignment and the CLIP Paradigm 5
2.2.2 Generative VLMs and Open-Weight Ecosystems 5
2.2.3 Medical Vision-Language Models 6
2.3 Decoders and Parameter-Efficient Fine-Tuning 8
2.3.1 Encoder-Decoder Architectures vs. Causal Modeling 8
2.3.2 Parameter-Efficient Fine-Tuning (PEFT) 9
2.4 Radiology Report Generation 10
2.4.1 Traditional and Early Transformer Approaches 10
2.4.2 The Evaluation Paradigm: From N-Grams to Clinical Efficacy . . 11
2.4.3 2D Radiology Foundation Models 11
2.4.4 3D Volumetric Report Generation 12
2.5 3D Medical Image Segmentation and Organ-Guided Attention 13
2.5.1 Volumetric Segmentation Frameworks 14
2.5.2 Organ-Guided Attention and Clinical Grounding 14

3 Methodology **16**
3.1 LLM-based Report Decomposition 16
3.1.1 Anatomical Schema Definition 16
3.1.2 Decomposition Pipeline 16

4 Experiments and Results **18**
4.1 Dataset 18
4.1.1 Dataset Overview 18
4.1.2 Preprocessing 18
4.2 Evaluation Metrics 19
4.2.1 Surface-level (Natural Language Generation) Metrics 20
4.2.2 Semantic / Embedding-based Metrics 21
4.2.3 Clinical / Structured Metrics 21
4.3 Results 23
4.3.1 Quantitative Results 23
4.3.2 Qualitative Results 23

5 Discussion **24**
5.1 Section 24
5.1.1 Subsection 24

6 Conclusion and Future Work **25**
6.1 Section 25
6.1.1 Subsection 25
6.2 Future Work 25

Abbreviations **26**

List of Figures **27**

List of Tables **28**

Bibliography **29**

1 Introduction

As deep learning has continued to advance over recent years, it has been applied to various fields of science and engineering. One of the most prominent applications of deep learning is in the field of computer vision. Deep learning models have achieved state-of-the-art performance on various computer vision tasks, such as image classification, object detection, and image segmentation. However, deep learning models are data-hungry and require large amounts of labeled data to achieve high performance. This is especially true for medical imaging, where obtaining large-scale labeled datasets is challenging due to privacy concerns and the high cost of expert annotation.

1.1 Problem Statement

Traditional NLP-based frameworks for radiology report generation often suffered from a lack of localized visual reasoning and a weak integration of clinical knowledge. While early 2D Vision-Language Models (VLMs) leveraging attention mechanisms improved the alignment between image regions and text tokens, they fundamentally failed to capture true three-dimensional anatomical context. In clinical practice, critical findings—such as lesion shape, spatial extent, and inter-slice continuity—depend entirely on volumetric data.

Applying modern VLM architectures directly to 3D medical volumes (like CT scans) introduces massive computational bottlenecks. The cubic growth of 3D attention mechanisms requires innovative bridging architectures, such as perceptual resamplers or Q-Formers, to compress thousands of spatial tokens into a dense semantic representation without overwhelming the language decoder. Furthermore, while the field of generative AI has seen an explosion of highly capable, multi-billion parameter foundation models, it remains an open question whether the "intelligence" driving accurate medical report generation lies primarily in the robust 3D vision encoder, or explicitly in the semantic reasoning capabilities of the Large Language Model decoder.

1.2 Thesis Formulation

This thesis addresses these gaps by evaluating state-of-the-art 3D Vision-Language Modeling techniques for radiology report generation. Specifically, we investigate whether extracting single, highly semantic visual tokens per anatomical organ provides a cleaner signal to the LLM decoder than complex, multi-token representations.

To achieve this, we leverage an organ-guided decomposition methodology where reports are broken down anatomical region by region, enabling the VLM to perform targeted, localized reasoning. Our experiments are anchored on the **CT-RATE** dataset, the first large-scale publicly available dataset pairing full 3D chest CT volumes with free-text radiology reports.

Instead of relying solely on traditional natural language metrics (BLEU, ROUGE) which often fail to capture clinical nuances—such as severely penalizing negative findings—this work assesses model performance using advanced clinical efficacy metrics like **RadGraph** and **GREEN** (Generative Radiology Report Evaluation and Error Notation), ensuring generated reports are clinically actionable rather than linguistically mimetic.

By running extensive ablations across decoder architectures—comparing massive, medically-aligned causal language models (e.g., **MedGemma**) against smaller generic causal models (**GPT-2**) and encoder-decoder architectures (**BioBART**)—this research uncovers the optimal architectural configurations for mapping sparse 3D visual tokens into structured clinical text.

1.3 Contributions

The main contributions of this thesis are as follows:

1. **Comprehensive 3D VLM Benchmarking**: We conduct one of the first extensive comparative analysis of decoder architectures (causal vs. encoder-decoder) coupled with 3D vision encoders for volumetric report generation.

2. **Organ-Guided Tokenization Strategy**: We demonstrate that compressing high-density 3D spatial features into a single, organ-specific visual token (via cross-attention) drastically outperforms overly complex, multi-token bridging architectures.

3. **Clinical Efficacy Prioritization**: We validate generated results strictly against clinical-factuality metrics (GREEN, RadGraph) to isolate true clinical reasoning from rote n-gram memorization.

2 Related Work

This chapter reviews the key areas of prior work that form the foundation for this thesis. Section 2.1 discusses the Vision Transformer architecture and its adaptation to medical imaging. Section 2.2 covers vision–language models, including both general-purpose and medical variants. Section 2.3 introduces large language models and parameter-efficient fine-tuning techniques. Section 2.4 surveys radiology report generation methods. Finally, Section 2.5 covers 3D medical image segmentation and organ-guided attention mechanisms.

2.1 Vision Transformers in Medical Imaging

The Vision Transformer (ViT) [1] fundamentally shifted the paradigm of computer vision by applying the transformer architecture, originally designed for natural language processing, to image recognition. ViT divides an input image into fixed-size non-overlapping patches, projects each patch into an embedding, and processes the resulting sequence through a standard transformer encoder. By treating image patches as tokens and relying entirely on multi-head self-attention, ViT eliminates the need for convolutional inductive biases (such as translation invariance and local receptive fields) and achieves competitive or superior performance compared to Convolutional Neural Networks (CNNs) when trained on massive datasets like ImageNet.

2.1.1 Architectural Evolutions: Hierarchical and Self-Supervised ViTs

While the original ViT required large-scale supervised pretraining, subsequent architectures addressed its limitations. The Swin Transformer [2] introduced a hierarchical structure with shifted windows, computing self-attention locally within non-overlapping windows that shift across layers. This hierarchical design reintroduces some useful inductive biases, making it highly efficient for dense prediction tasks like segmentation and object detection, which are critical in medical imaging contexts where precise anatomical boundaries matter.

Simultaneously, advancements in self-supervised learning alleviated the reliance on massive labeled datasets—a severe bottleneck in the medical domain. Masked Autoencoders (MAE) [3] proposed an asymmetric encoder-decoder architecture where the

encoder operates only on an unmasked subset of patches (e.g., 25%), and a lightweight decoder reconstructs the heavily masked (75%) original image. This approach enables the learning of robust, high-level visual representations directly from unlabeled data. Building upon self-supervised paradigms, DINOv2 [4] demonstrated that models trained via self-distillation with no labels can learn universal visual features that are highly transferable across diverse domains, capturing part-level semantic correspondences without explicit supervision.

2.1.2 Adaptation to 3D Medical Volumes

In the medical imaging domain, transformers have been widely adopted to capture the long-range dependencies crucial for understanding complex anatomical structures. Early adaptations like TransUNet [5] combined a CNN encoder for local feature extraction with a transformer bottleneck for global context in 2D medical image segmentation.

However, clinical data such as Computed Tomography (CT) and Magnetic Resonance Imaging (MRI) are inherently 3D volumetric scans. Processing these volume slice-by-slice discards critical through-plane spatial context. UNETR [6] addressed this by using a ViT encoder to process 3D volumetric data directly, extracting 3D patches from the input volume.

The transition from 2D image patches to 3D volumetric patches introduces a catastrophic computational bottleneck. In a standard self-attention mechanism, memory and computational complexity grow quadratically with the sequence length. For a 3D medical image, the sequence length expands cubically with respect to the spatial dimensions. If a high-resolution CT scan is embedded into thousands or even tens of thousands of 3D tokens, global self-attention rapidly exceeds the memory limits of modern GPUs. To mitigate this, state-of-the-art 3D Vision Transformers increasingly rely on sparse attention mechanisms, shifted window paradigms (as seen in 3D-Swin variants), or aggressive spatial downsampling schemes early in the network hierarchy.

Further innovations in 3D transformer architectures have focused on efficient upsampling and token processing. For instance, techniques like Pixel Shuffle [7], originally popular in super-resolution networks, have been ingeniously adapted to 3D transformer upsampling. By reshaping channel dimensions into spatial dimensions, 3D Pixel Shuffle provides an extremely compute-efficient method to reconstruct high-resolution 3D medical images from heavily compressed token bottleneck representations, preserving spatial topologies without the massive parameter overhead of transposed 3D convolutions.

The core vision encoder used in this thesis is a 3D ViT initialized with MAE-pretrained weights. Unlike standard 2D ViTs, this encoder processes volumetric CT inputs using 3D patch embeddings, meticulously preserving spatial relationships across the axial,

coronal, and sagittal planes. By leveraging masked autoencoding objectives during pretraining, the encoder learns to prioritize clinically salient structural boundaries even before fine-tuning, enabling true, robust volumetric reasoning for downstream vision-language tasks.

2.2 Vision–Language Models

Vision–language models (VLMs) aim to bridge the semantic gap between visual perception and natural language understanding. These models learn joint representations of images and text, enabling a wide array of cross-modal tasks such as image captioning, visual question answering (VQA), and complex multi-modal reasoning.

2.2.1 Contrastive Alignment and the CLIP Paradigm

CLIP (Contrastive Language–Image Pre-training) [8] introduced a highly scalable approach to learning visual representations from natural language supervision. By training independent vision and text encoders to maximize the cosine similarity of matching image–text pairs in a shared embedding space while minimizing the similarity of millions of non-matching pairs, CLIP learns highly transferable visual concepts. Trained on 400 million image–text pairs curated from the internet, CLIP demonstrates unprecedented zero-shot transfer capabilities to a wide range of downstream visual tasks.

Recent advances have refined the CLIP objective. SigLIP (Sigmoid Loss for Language Image Pre-Training) [9] replaces the standard softmax-based contrastive loss with a pairwise sigmoid loss. This seemingly simple change eliminates the need for global views of all negative pairs across devices, dramatically improving training efficiency and scalability, allowing for massive batch sizes and yielding stronger representations. The contrastive learning paradigm established by CLIP and optimized by SigLIP has become the bedrock foundational building block for aligning vision and language modalities in modern VLMs.

2.2.2 Generative VLMs and Open-Weight Ecosystems

While contrastive models focus purely on alignment and retrieval, generative VLMs extend the paradigm to produce natural language outputs explicitly conditioned on visual inputs. BLIP (Bootstrapping Language-Image Pre-training) [10] proposed a unified framework that jointly trains a contrastive alignment objective, an image–text matching classification objective, and a generative language modeling objective. This

multi-task design enables BLIP to be highly effective for both comprehensive image understanding and robust text generation tasks.

Building upon this, BLIP-2 [11] introduced a highly compute-efficient strategy for bridging the modality gap: the Q-Former (Query Transformer). The Q-Former acts as an informational bottleneck between a frozen image encoder and a frozen large language model. It utilizes a set of learnable query vectors that cross-attend to the dense visual features extracted by the image encoder, forcing the queries to extract only the most text-relevant visual information. By doing so, the Q-Former drastically compresses thousands of spatial visual tokens into a fixed, small number of highly semantic latent tokens (e.g., 32 tokens). These compressed tokens are then linearly projected directly into the text embedding space of the frozen LLM. This architectural breakthrough demonstrated that one can achieve state-of-the-art multi-modal capabilities without unfreezing the massive weights of the underlying language or vision foundation models, saving enormous computational resources.

A similar design philosophy for handling high-density visual signals was famously adopted by DeepMind's Flamingo [12]. Rather than a Q-Former, Flamingo utilized a Perceiver Resampler mechanism, which also cross-attends to a massive input array (such as a high-resolution image or a varying-length video) using a small set of latent, learnable query vectors. Flamingo then interleaved these compressed visual tokens with standard text tokens inside a frozen LLM through gated cross-attention layers, demonstrating unparalleled few-shot learning and in-context reasoning capabilities.

LLaVA (Large Language-and-Vision Assistant) [13] vastly simplified these complex bridging architectures by proving that a straightforward Multi-Layer Perceptron (MLP) projection layer could effectively map visual features directly from a pretrained CLIP encoder into the native vocabulary embedding space of an LLM, entirely discarding the need for Q-Formers or Resamplers. Through visual instruction tuning—training the model on instruction-following datasets combining images with conversational text—LLaVA achieved remarkable multi-modal dialogue capabilities. Recently, open-source initiatives have rapidly accelerated VLM research. Models like InternVL [14] focus on massive scaling of the vision foundation model to better align with generic linguistic tasks, while the Molmo family [15] emphasizes the crucial role of highly curated, completely open-source point-based multimodal data to achieve state-of-the-art results with highly efficient projector architectures.

2.2.3 Medical Vision-Language Models

Adapting VLMs to the medical domain presents significant unique challenges. Medical images (such as high-resolution multi-parametric MRIs or chest X-rays) have fundamentally different characteristics from natural images (like those in COCO/ImageNet).

Furthermore, clinical texts utilize a dense, highly specialized vocabulary, and the margin for error in medical AI is essentially zero.

Early attempts to bridge this gap focused on domain-specific contrastive pretraining. BiomedCLIP [16] addressed this by pretraining a CLIP-style architecture from scratch on 15 million biomedical image–text pairs sourced from PubMed Central articles, demonstrating that in-domain pretraining drastically improves performance on downstream biomedical classification and retrieval tasks compared to generic CLIP models.

For 2D clinical radiology, the development of foundation models has seen rapid acceleration [17]. CheXagent [18] was introduced as a clinical foundation model for chest X-ray interpretation, fine-tuned specifically on massive corpora of radiology reports to both interpret images and answer complex clinical questions accurately. MAIRA-2 [19] pushed the state-of-the-art further by introducing grounded radiology report generation, forcing the model to explicitly link generated clinical assertions in the text back to localized bounding boxes in the original X-ray, drastically reducing hallucinations and improving clinical trustworthiness.

For 3D medical imaging, the challenge is compounded by the sheer size of the volumetric data and the scarcity of paired text data. Hamamci et al. [20] introduced CT-CLIP, a contrastive learning framework specifically engineered for 3D computed tomography. Trained exclusively on the massive CT-RATE dataset—which pairs unannotated 3D volumetric CT scans directly with their corresponding free-text radiology reports—CT-CLIP successfully learns 3D structural visual representations that map directly to the clinical text manifold. This foundational work represents one of the first successful large-scale efforts to apply multi-modal vision–language pretraining directly to unstructured volumetric medical data.

MedGemma [21] is a family of medical foundation models built on top of Google's rigorous Gemma architecture [22]. MedGemma structurally inherits the robust capabilities of the general-purpose Gemma model but undergoes rigorous continued pretraining and specialized instruction-tuning on vast troves of verified biomedical literature, medical board exams, and de-identified clinical notes. The resulting instruction-tuned variant (MedGemma-4B-IT) demonstrates exceptional zero-shot and few-shot capability on complex medical reasoning tasks, making it a highly compelling, medically-aligned language decoder for complex medical VLM pipelines, and ultimately serving as one of the core generative components evaluated in this thesis.

2.3 Decoders and Parameter-Efficient Fine-Tuning

The generation of complex radiology reports from visual features requires a demonstrably powerful language decoder. The development of natural language generation models has largely bifurcated into two fundamentally distinct architectural paradigms: Encoder-Decoder models and Causal Language Models. Understanding the mechanistic differences between these paradigms is crucial when selecting a backbone for medical vision-language reasoning.

2.3.1 Encoder-Decoder Architectures vs. Causal Modeling

Encoder-Decoder models, such as BART (Bidirectional and Auto-Regressive Transformers) [23] and T5, utilize a bidirectional encoder to process the input context comprehensively and an autoregressive decoder to generate the output sequence token by token. Because the encoder can attend to both left and right context simultaneously during the prompt processing phase, these models exel at tasks requiring deep semantic understanding of the entire input condition before generating text (e.g., summarization or machine translation). BART is specifically pretrained by corrupting documents (via token masking, deletion, and sentence permutation) and optimizing a reconstruction loss.

In the biomedical domain, BioBART [24] adapts this exact architecture by continually pretraining the baseline BART model on a massive corpus of PubMed abstracts. This domain-specific pretraining allows BioBART to inherit the robust grammatical generation capabilities of BART while rigidly aligning its vocabulary and internal latent representations with complex, highly specialized medical terminology. In standard vision-language pipelines utilizing Encoder-Decoder backbones, visual embeddings are typically fed directly into the bidirectional encoder, allowing the system to build a rich, globally aware contextual representation before the autoregressive generation phase even begins.

Conversely, Causal Language Models (CLMs) utilize a decoder-only architecture that predicts the next token in a sequence strictly based on preceding leftward tokens. GPT-2 (Generative Pre-trained Transformer 2) [25] popularized this highly scalable paradigm, demonstrating that simple unsupervised next-token prediction on diverse web corpora could intuitively yield models capable of coherent, zero-shot, open-ended text generation. Due to its relatively modest scale (e.g., 124M parameters for the base model) and exceptionally clean architectural foundation, GPT-2 remains a popular, tractable baseline for experimental diagnostic text generation tasks where compute is constrained.

Unlike bidirectional encoders, causal models process the conditioning prompt (includ-

ing visual tokens prepended as a "soft prompt") autoregressively. While theoretically limiting the deep bidirectional mingling of context tokens, the massive parameter count of modern causal models overrides this limitation through sheer representational capacity.

Recent foundation models have scaled the decoder-only paradigm to tens of billions of parameters. The LLaMA family [26] and the Gemma family [22] have demonstrated unprecedented capabilities in generalized natural language understanding, logical reasoning, and complex instruction-following. MedGemma [21], a specially curated medical variant (e.g., MedGemma-4B-IT), extends these immense capabilities by injecting rigorously structured medical knowledge into the weights during continued pretraining. By fine-tuning on vast repositories of de-identified clinical notes and structured medical questionnaires, MedGemma serves as a state-of-the-art causal decoder. In causal VLM architectures, preserving the spatial and semantic integrity of the visual token prefix is paramount, as the entire subsequent automated text generation hinges upon the initial visual context.

2.3.2 Parameter-Efficient Fine-Tuning (PEFT)

Adapting massive decoders (especially multi-billion parameter foundation models like MedGemma) to specialized downstream tasks such as highly localized, organ-specific radiology report generation via full fine-tuning is computationally prohibitive and often leads to catastrophic forgetting of the base model's general linguistic capabilities. Parameter-efficient fine-tuning (PEFT) methods meticulously address this by updating only a tiny fraction of the model's parameters while keeping the vast majority of the expensive pretrained weights strictly frozen.

Early PEFT strategies included Adapter modules [27], which introduced small, trainable feed-forward bottleneck networks sandwiched between the frozen multi-head attention and feed-forward transformer sub-layers. Another approach, Prefix-Tuning [28], mathematically optimizes a sequence of continuous task-specific prompt vectors rigidly prepended to the transformer layers. While effective in mitigating catastrophic forgetting, these methods fundamentally alter the network topology, often introducing undesirable inference latency or artificially restricting the available context window depth.

LoRA (Low-Rank Adaptation) [29] dynamically elegantly sidestepped these issues, emerging as the overwhelmingly dominant PEFT method for modern foundation models. LoRA theoretically hypothesizes that the necessary updates to the dense weight matrices during specialized fine-tuning possess a profoundly low intrinsic rank. It introduces fully trainable low-rank decomposition matrices directly in parallel with the linear layers of the frozen model (most commonly targeting the query and

value projection matrices within the self-attention mechanism). Instead of massively updating the dense, full weight matrix $W_0 \in \mathbb{R}^{d \times k}$, LoRA learns two exponentially smaller matrices $A \in \mathbb{R}^{d \times r}$ and $B \in \mathbb{R}^{r \times k}$ where the rank $r \ll \min(d, k)$, such that the adapted functional weight becomes $W_0 + \Delta W = W_0 + BA$. Crucially, because the matrix product BA can be structurally merged directly back into W_0 prior to model deployment, LoRA introduces absolutely zero inference latency overhead.

To further democratize and accelerate training, QLoRA [30] was introduced, harmonizing LoRA with advanced 4-bit Normal Float (NF4) double quantization techniques. In QLoRA, the massive frozen pretrained model weights reside efficiently in ultra-low 4-bit precision, while only the lightweight, trainable LoRA adapter matrices are temporarily cast to higher precision (e.g., BFloat16) during the backpropagation pass. This groundbreaking combination effectively slashed the VRAM requirements for fine-tuning multi-billion parameter foundation models, allowing researchers to rapidly iterate and optimally adapt models like MedGemma on standard consumer-grade hardware architectures, making the development of highly advanced, computationally constrained medical VLMs broadly accessible.

2.4 Radiology Report Generation

Automatic radiology report generation is the task of producing clinically accurate free-text reports from medical images. This task is fundamentally more challenging than generic image captioning because generated reports must be factually accurate, cover all relevant pathological findings (and explicitly note the absence of others), and strictly adhere to dense clinical conventions.

2.4.1 Traditional and Early Transformer Approaches

Early approaches to radiology report generation adapted encoder–decoder architectures directly from the natural image captioning literature. Jing et al. [31] proposed a co-attention mechanism combined with a hierarchical LSTM decoder to generate structured medical reports from chest X-rays. However, these CNN–RNN frameworks typically produced generic, repetitive descriptions and struggled significantly with capturing rare or subtle findings, often suffering from severe data imbalance in clinical datasets.

Transformer-based methods subsequently improved report generation quality by better tracking long-term context. Chen et al. [32] introduced a memory-driven transformer that used a relational memory module to record key clinical information during autoregressive generation, markedly improving the coherence of long reports. In a follow-up, Chen et al. [33] proposed cross-modal memory networks that maintained separate but interacting memory banks for visual and textual features, enabling more

robust cross-modal reasoning. Demonstrating the value of initialization, Nicolson et al. [34] showed that warm-starting both the vision encoder (e.g., from ImageNet or CheXpert) and the language decoder (e.g., from clinical BERT) substantially improved chest X-ray report generation over training from scratch.

2.4.2 The Evaluation Paradigm: From N-Grams to Clinical Efficacy

A foundational obstacle in early radiology report generation research was the over-reliance on traditional Natural Language Generation (NLG) metrics originally designed for machine translation—such as BLEU, ROUGE, and METEOR. These metrics inherently quantify lexical overlap; they measure how many identical n-grams the generated text shares with the ground truth. While effective for general captioning, they are critically flawed in the medical domain. A single word substitution (e.g., swapping "no" for "a") has a minimal penalty on a BLEU score but completely reverses the clinical diagnosis of the report, potentially causing severe harm in a deployed setting.

To address this critical gap, the research community underwent a massive paradigm shift toward metrics that explicitly measure clinical efficacy and semantic correctness. Early clinical metrics like CheXbert [35] leveraged an NLP labeler to extract predefined clinical observations (e.g., Pneumonia, Cardiomegaly) from both the generated and target reports, scoring the model based on diagnostic accuracy rather than exact wording.

Subsequent innovations focused on capturing the complex relational structure of medical language. The RadGraph metric [36] evaluates reports by parsing them into intricate knowledge graphs (nodes representing entities like anatomy or observation, and edges representing relations like "located at" or "modify"), computing the overlap between these highly structured clinical graphs. More recently, leveraging the advent of LLMs, the GREEN (Generative RadiOlogy Report Evaluation and Error aNnotation) metric provides both a quantitative clinical score and qualitative feedback, closely aligning with expert radiologist judgments. This shift in evaluation metrics forced a corresponding shift in architectural design: models could no longer rely on memorizing high-frequency text templates to game lexical scores; they were required to genuinely perceive and report the underlying pathology.

2.4.3 2D Radiology Foundation Models

As the field moved toward the foundation model paradigm, researchers began adapting massive, frozen large language models as report decoders. R2GenGPT [37] demonstrated that a frozen LLM combined with a trainable visual encoder and a lightweight projection layer could generate high-quality radiology reports that performed exceed-

ingly well on both NLG and clinical efficacy metrics. This paradigm benefits directly from the immense language understanding capabilities and medical knowledge embedded in pretrained LLMs while keeping multi-modal training costs manageable.

Recently, the field has seen the emergence of true foundation models specifically tailored for 2D X-ray interpretation. CheXagent [18] stands out as a pioneering clinical foundation model designed to interpret chest X-rays. By fine-tuning a massive VLM on diverse corpora of radiology reports and visual question-answering pairs, CheXagent achieved state-of-the-art performance in both free-text report generation and interactive clinical dialogue.

Simultaneously, the architectural focus shifted from mere linguistic fluency to rigorous clinical grounding. MAIRA-2 [19] introduced the concept of grounded radiology report generation. Instead of hallucinating plausible text based on global image context, MAIRA-2 explicitly maps generated clinical assertions in the text back to localized bounding boxes in the original X-ray. This spatial grounding dramatically reduces clinical hallucinations—a critical safety requirement for medical deployment.

2.4.4 3D Volumetric Report Generation

While 2D X-ray interpretation has matured rapidly, 3D volumetric report generation (e.g., from CT or MRI scans) remains an open and highly challenging frontier. The jump from 2D to 3D exponentially increases the computational complexity of the visual token sequence. Furthermore, until the recent release of the CT-RATE dataset [20], there was a severe scarcity of paired 3D volumes and free-text reports available for training.

The release of CT-RATE spurred the rapid development of several pioneering 3D multimodal foundation models:

- **CT2Rep** [38]: One of the earliest works to leverage the CT-RATE dataset, CT2Rep demonstrated that automated radiology report generation for 3D medical imaging is feasible by properly projecting 3D visual tokens into a robust language decoder.
- **M3D** [39]: Advancing 3D medical image analysis with Multi-Modal Large Language Models, M3D proposed strategies to efficiently compress volumetric data into spatial tokens, enabling the LLM to process entire CT scans without exceeding context windows.
- **Argus** [40]: Argus introduced comprehensive benchmarking and enhancement strategies for VLMs in 3D radiology, testing models ranging from 3B configurations to massive 70B parameter deployments. Argus highlighted that scaling the LLM capacity yields direct improvements in clinical efficacy metrics like GREEN.

- **Med3DVLM** [41] and **RadFM** [42]: Both models established unified 3D vision-language frameworks, focusing on cross-modal alignment between highly unstructured 3D voxel spaces and precise clinical text.

A key challenge unifying these works is token compression. Because a raw 3D CT volume contains vastly more information than a 2D image, naive patch extraction results in tens of thousands of tokens—overwhelming the LLM decoder. Recent studies [43] emphasize that advanced tokenization strategies (such as Perceiver-based latent resampling, spatial 3D pooling, or learned region-of-interest selection) are absolutely critical for effective 3D VLM performance.

Table 2.1: Performance benchmarking of recent 3D Volumetric Foundation Models for Radiology Report Generation evaluated on the CT-RATE dataset. Metrics highlight the paradigm shift toward clinical efficacy (GREEN) over traditional NLP metrics (BLEU-4). Data adapted from recent multi-modal surveys.

Model Name	Parameters	Input Resolution	BLEU-4	ROUGE-L	METEOR	GREEN
Argus-70B	70B	$512 \times 512 \times 256$	47.98	-	-	**60.22**
Argus-8B	8B	$512 \times 512 \times 256$	45.81	-	-	58.42
Argus-3B	3B	$512 \times 512 \times 256$	43.03	-	-	55.56
CT2Rep-3B	3B	$256 \times 256 \times 64$	38.77	48.96	-	51.72
M3D-3B	3B	$256 \times 256 \times 64$	40.23	50.31	-	50.31
RadFM-3B	3B	$256 \times 256 \times 64$	39.95	49.46	-	49.46
COLIPRI-CRM	1B	$160 \times 160 \times 160$	45.20	55.20	57.80	-

As shown in Table 2.1, there is a clear scaling law where higher resolution inputs and larger language decoders (e.g., Argus-70B) correlate non-linearly with higher clinical efficacy scores (GREEN).

The approach adopted in this thesis builds directly upon this frontier: a trainable 3D vision encoder (with variants exploring multi-scale and organ-guided context) produces visual embeddings that are projected specifically into the embedding space of a medically aligned MedGemma decoder, targeting the complex transition from global captioning to precise, localized, organ-specific 3D reporting.

2.5 3D Medical Image Segmentation and Organ-Guided Attention

Accurate anatomical localization is an essential prerequisite for generating highly specific, clinically actionable radiology reports. The availability of robust, automated

whole-body 3D segmentation tools has enabled novel VLM architectures that explicitly leverage anatomical priors to ground their text generation.

2.5.1 Volumetric Segmentation Frameworks

While early segmentation models relied on 3D CNNs (e.g., 3D U-Net), transformer-based architectures have increasingly dominated this space by effectively capturing long-range contextual dependencies. TransUNet [5] and Swin-UNet [44] demonstrated the efficacy of transformers for 2D medical segmentation, but volumetric data requires dedicated 3D processing.

nnFormer (Interleaved Transformer for Volumetric Medical Image Segmentation) [45] extends transformer capabilities explicitly to 3D medical images. By interleaving local 3D convolution operations with global multi-head self-attention mechanisms, nnFormer captures both the intricate local textures of tumors and the global geometrical relationships of large organs, setting strong benchmarks across multiple 3D medical datasets.

In practical clinical pipelines, TotalSegmentator [46] has emerged as an indispensable, widely used utility for automatic segmentation. Built upon the robust nnU-Net framework, TotalSegmentator provides highly reliable, out-of-the-box segmentation of 104 distinct anatomical structures—including major organs, bones, vessels, and muscles—from standard CT images. The discrete 3D masks produced by such algorithms can serve directly as precise anatomical priors for downstream generative tasks.

2.5.2 Organ-Guided Attention and Clinical Grounding

Incorporating these anatomical priors into VLM architectures enables a critical paradigm shift: from global image captioning to localized, organ-specific reasoning. Rather than processing an entire CT volume as a uniform sequence of visual tokens, organ-guided mechanisms restrict or bias the model's cross-attention to regions corresponding strictly to specific query structures (e.g., the lungs or the liver).

This approach draws heavily upon learnable query architectures such as the Perceiver [47], where a fixed set of latent vectors attends to a massive sensory input. In the medical domain, these latent vectors can represent specific organs. By masking the visual features using 3D segmentation maps prior to cross-attention, the language decoder is forced to focus only on the relevant anatomy when generating the corresponding sentence in the radiology report.

Furthermore, in the context of multi-task learning [48], organ-specific feature extraction can be significantly enhanced via auxiliary classification objectives. By jointly training the visual encoder to perform specific disease classification tasks alongside

open-ended report generation, the model learns a more disentangled and discriminative organ-level representation. This auxiliary grounding acts as a powerful inductive bias, encouraging the visual encoder to capture the subtle physiological abnormalities that drive accurate clinical reporting.

3 Methodology

3.1 LLM-based Report Decomposition

To enable fine-grained, organ-specific radiology report generation, we implement a preprocessing pipeline that decomposes full free-text radiology reports into structured, organ-specific findings. This is achieved using a Large Language Model (LLM) steered by a predefined anatomical schema.

3.1.1 Anatomical Schema Definition

We define a comprehensive schema covering major anatomical regions in chest CT scans. The schema targets the following structures:

- **Thorax**: Lung, Heart, Aorta, Trachea, Esophagus, Portal Vein, Rib.
- **Abdomen**: Liver, Gallbladder, Stomach, Pancreas, Spleen, Kidney, Adrenal, Colon, Small Bowel, Bladder.

This schema ensures that the model learns to associate specific image regions with their corresponding textual descriptions, rather than generating a monolithic report.

3.1.2 Decomposition Pipeline

We utilize the **Llama-3.3-70B-Instruct** model for the decomposition task. Due to the significant memory requirements of this 70-billion parameter model, we employ the **AWQ (Activation-aware Weight Quantization)** version [49] to fit the model within the memory constraints of our GPU infrastructure (e.g., 4×A100). The model is served using **vLLM** [50], a high-throughput and memory-efficient LLM serving engine.

The decomposition process involves the following steps:

1. **Input**: The raw "Findings" and "Impression" sections of the radiology report are concatenated.
2. **Prompting**: The LLM is prompted with a system instruction to act as a radiologist and extract organ-specific information according to our defined JSON schema.

The prompt explicitly requests valid JSON output and handles missing organs by assigning them a `null` value.

3. **Output Parsing**: The generated output is parsed to extract the JSON structure, separating findings and impressions for each organ.

The implementation uses a robust parsing mechanism to handle potential formatting irregularities in the LLM's output, ensuring high data quality for downstream training.

4 Experiments and Results

4.1 Dataset

We validate our methods using the **CT-RATE** dataset [20], a large-scale multimodal dataset for 3D computed tomography. CT-RATE is the first large-scale publicly available dataset that pairs full 3D medical volumes with free-text radiology reports, enabling the training of end-to-end 3D vision–language models for tasks such as report generation and text-conditional image synthesis [51].

4.1.1 Dataset Overview

The dataset consists of 25,692 3D chest CT volumes from 21,304 unique patients, paired with their corresponding free-text radiology reports. The data was collected from multiple sources and anonymized to ensure privacy. Key statistics of the dataset are summarized in Table 4.1 [20].

Table 4.1: Key statistics of the CT-RATE dataset used in our experiments.

Statistic	**Value**
Total CT Volumes	25,692
Unique Patients	21,304
Modality	3D Chest CT
Text Modality	Radiology Reports (Findings & Impression)

4.1.2 Preprocessing

Since CT volumes in the CT-RATE dataset vary significantly in all dimensions, we applied a standardized preprocessing pipeline to ensure a stable training input for our models. The steps, implemented using the MONAI library, are as follows:

1. **Intensity Scaling**: We apply distinct intensity scaling to normalize Hounsfield Units (HU) to the range $[0, 1]$. We clip intensities between -1150 and 350 HU,

which covers the relevant radiodensity range for lung and soft tissue structures while suppressing extreme outliers.

2. **Spatial Normalization**: To handle the variable volume sizes, we standardize the spatial dimensions to a fixed size of $112 \times 256 \times 352$ voxels. This is achieved by first applying *Spatial Padding* with constant zero values to ensure the volume is at least the target size.

3. **Dimension Reordering**: We ensure a channel-first data layout and transpose dimensions to match the input expectations of our 3D vision encoder.

- **Text Preprocessing**: To enable organ-level report generation, we employed an LLM-based decomposition strategy. We used a Large Language Model to decompose each full radiology report into organ-specific descriptions (e.g., "Lung", "Heart", "Liver"). This allows our model to learn fine-grained section-specific representations. We specifically filter for and utilize these decomposed sections, tokenizing them with a maximum sequence length of 128 tokens for our decoder.

Recent work has shown that using improved tokenization strategies can further enhance performance on this dataset [43]. We utilize the official train/validation/test splits provided by the authors to ensure comparable results with prior work.

4.2 Evaluation Metrics

To assess the performance of our radiology report generation system we evaluate generated reports with multiple complementary metrics. Metrics are grouped by the type of signal they capture:

1. **Surface-level (linguistic) metrics:** measure n-gram overlap and surface similarity to reference reports.

2. **Semantic / embedding-based metrics:** measure semantic similarity using contextual embeddings.

3. **Clinical / structured metrics:** evaluate clinical correctness by extracting findings, entities, relations or clinically-relevant labels.

4. **Human evaluation and statistical tests:** measure radiologist preference, clinical usefulness, and statistical significance.

4.2.1 Surface-level (Natural Language Generation) Metrics

These metrics are commonly used in image captioning and report generation benchmarks and capture lexical and syntactic similarity.

BLEU

BLEU [52] is an n-gram precision-based metric. For a candidate report c and a set of references $\{r\}$, BLEU is computed as

$$\text{BLEU} = \text{BP} \cdot \exp\left(\sum_{n=1}^{N} w_n \log p_n\right), \tag{4.1}$$

where p_n is the modified precision for n-grams, w_n are weights (we use uniform weights $w_n = 1/N$), and BP is the brevity penalty:

$$\text{BP} = \begin{cases} 1 & \text{if } c_{len} > r^*_{len} \\ \exp\left(1 - \frac{r^*_{len}}{c_{len}}\right) & \text{otherwise} \end{cases} \tag{4.2}$$

Here c_{len} is the candidate length and r^*_{len} is the effective reference length (e.g., closest reference length). We report BLEU-4.

ROUGE-L

ROUGE-L [53] uses the length of the Longest Common Subsequence (LCS) to capture sentence-level structure. Define

$$R_{LCS} = \frac{\text{LCS}(c, r)}{|r|}, \qquad P_{LCS} = \frac{\text{LCS}(c, r)}{|c|},$$

then the ROUGE-L F-measure is

$$F_{LCS} = \frac{(1 + \beta^2)\, R_{LCS}\, P_{LCS}}{R_{LCS} + \beta^2 P_{LCS}}, \tag{4.3}$$

where β weights recall relative to precision. In practice we set $\beta = 1$ to report the balanced F_1 unless otherwise specified.

METEOR

METEOR [54] aligns unigrams and computes a weighted harmonic mean of precision and recall that emphasizes recall. The core F_{mean} is:

$$F_{\text{mean}} = \frac{(1 + \alpha) PR}{R + \alpha P}, \tag{4.4}$$

where typical METEOR uses $\alpha = 9$ (equivalently $F_{\text{mean}} = \frac{10PR}{R+9P}$). METEOR applies a fragmentation penalty:

$$\text{Penalty} = 0.5 \left(\frac{\#\text{chunks}}{\#\text{matches}} \right)^3, \tag{4.5}$$

and final score:

$$\text{METEOR} = F_{\text{mean}} \cdot (1 - \text{Penalty}). \tag{4.6}$$

CIDEr

CIDEr [55] uses TF–IDF weighting for n-grams and computes average cosine similarity between a candidate's and references' TF–IDF vectors. For n-grams of length n:

$$\text{CIDEr}_n(c, r) = \frac{1}{M} \sum_{i=1}^{M} \frac{\mathbf{g}^n(c) \cdot \mathbf{g}^n(r_i)}{\|\mathbf{g}^n(c)\| \, \|\mathbf{g}^n(r_i)\|}, \tag{4.7}$$

and final CIDEr is the (optionally IDF-normalized) average across $n = 1 \ldots 4$.

Notes: Surface metrics are sensitive to wording and may penalize clinically-correct paraphrases. Use them together with semantic and clinical metrics.

4.2.2 Semantic / Embedding-based Metrics

These metrics capture deeper (contextual) semantic similarity beyond n-gram overlap.

BERTScore

BERTScore computes pairwise token similarities using contextual token embeddings from a pre-trained transformer (e.g., BERT). For tokens in candidate and reference, token-wise cosine similarities are computed and aggregated into precision, recall and an F_1 score; optional IDF weighting can be applied [56].

BERT-based contextual similarity

Besides BERTScore, one can compute sentence-level embeddings (CLS / pooled embeddings) and cosine similarity, or use more specialized biomedical encoders (e.g., BioBERT, ClinicalBERT) to increase domain alignment.

4.2.3 Clinical / Structured Metrics

These metrics aim to measure clinical correctness and the presence/absence of findings.

CheXbert

CheXbert [35] is a BERT-based labeler trained on the output of the CheXpert rule-based labeler to extract findings from radiology reports with higher accuracy. We use it to detect the presence of 14 standard observations (e.g., Cardiomegaly, Edema, Pneumonia, No Finding). We compute the Micro-F_1 score across all 14 classes to measure the clinical accuracy of the generated reports compared to the ground truth.

RadGraph

RadGraph [36] converts radiology reports into knowledge graphs (entities and relations). Evaluation is done by measuring overlap between predicted and reference graphs; typical metrics are node (entity) Precision/Recall/F_1 and edge (relation) Precision/Recall/F_1. Reporting both entity-level and relation-level scores helps diagnose whether models miss or hallucinate links between findings and anatomy.

SRR-BERT / Structured-report metrics

SRR-BERT and similar methods fine-tune BERT-like encoders to score structured-report correctness; they are useful for fine-grained disease classification and structured-report components. Report the metric(s) as provided by the original implementation (precision/recall/F_1 or regression correlation).

GREEN and other factuality metrics

GREEN (Generative Radiology report Evaluation and Error Notation) and related metrics are designed to highlight factual errors and categorize them by severity (critical vs non-critical). Use them to count clinically-significant hallucinations and factual inconsistencies.

Composite clinical metrics (e.g., RadCliQ)

Composite metrics such as RadCliQ combine surface (e.g., BLEU) and clinical (e.g., CheXpert) signals, often with learned weights to better correlate with radiologist judgments. When using a learned composite, report how weights were obtained and validate correlation with human raters.

4.3 Results

4.3.1 Quantitative Results

Content goes here.

4.3.2 Qualitative Results

Content goes here.

5 Discussion

5.1 Section

Content goes here.

5.1.1 Subsection

More content goes here.

6 Conclusion and Future Work

6.1 Section

Content goes here.

6.1.1 Subsection

More content goes here.

6.2 Future Work

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Abbreviations

List of Figures

List of Tables

2.1 Performance benchmarking of recent 3D Volumetric Foundation Models for Radiology Report Generation evaluated on the CT-RATE dataset. Metrics highlight the paradigm shift toward clinical efficacy (GREEN) over traditional NLP metrics (BLEU-4). Data adapted from recent multi-modal surveys. 13

4.1 Key statistics of the CT-RATE dataset used in our experiments. 18

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